

Current-Control Circuit for LED Lighting



YOGESH SHUKLA

LED is basically a low-voltage device, which requires specified DC current and voltage across it for its operation. It can adapt to different power sources. Here is a circuit for controlling the current in LED lamps without using a decoupling capacitor at the supply voltage. This circuit uses an LED driver instead.

The main problem that affects the working of low-voltage LED lamps, with a chopped power supply, is the

continuous charging and discharging of the decoupling capacitor. By switching off the LED when the supply voltage is low, discharging of capacitor can be avoided if an LED driver is used. The driver protects the LED from input voltage fluctuation by providing a constant output for the operation.

Circuit and working

Fig. 1 shows the circuit diagram of the current-controlled circuit for a low-voltage LED lighting system. It is built around MAX16832 (IC1), six 1-watt white LEDs (LED1 through LED6), transistors BC557 (T1) and BC547 (T2), and few other components.

LED driver MAX16832 used in the circuit has dedicated DIM input (pin 7) for switching the LED current. Signal to drive DIM input is extracted

from the LED driver circuit. This circuit detects the starting of the off-time and switches off the LED before the decoupling capacitor is heavily discharged. The driver circuit switches on the LED when the supply voltage is high. The dimming capability of the driver can dim the LED output up to 100 per cent.

During its on time, the base of transistor T1 gets a positive voltage. This drives pnp transistor T1 into off state. Since there is no current in R2, transistor T1's collector will be at zero volts. Transistor T2, being an npn transistor, will also be in off state. Transistor T2 and resistors R3 and R4 form an inverter, which converts the logic zero at collector of T1 to logic one at collector of T2. This turns the LEDs on via DIM pin of IC1.

As soon as the off time starts, the input voltage drops rapidly, but the envelope detector comprising diode D1 and capacitor C2 reacts slowly. The base voltage at T1 drops faster than its emitter voltage. T1 switches on when -0.7 volts is achieved at its base-emitter. This changes

Test Points

Test point	Details
TP0	GND (reference for measurement)
TP1	24V
TP2	10kHz to 100kHz when S2 is at position 1

PARTS LIST

Semiconductors:

- IC1 - MAX16832 LED driver IC
- T1 - BC557 pnp transistor
- T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2, R4 - 100-kilo-ohm
- R3 - 2-kilo-ohm
- R5 - 220-ohm
- D1 - 1N4007 rectifier diode
- D2 - 1N5819 schottky diode
- LED1-LED6 - 1-watt white LED

Capacitors:

- C1, C2 - 1 μ F, 50V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 2-pin connector
- S1 - On/off switch
- S2 - 2-way rotary switch
- 24V DC power supply

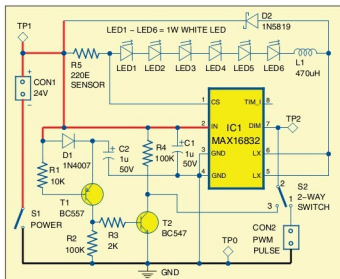


Fig. 1: Circuit diagram of the current-control circuit for an LED lighting system



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